



RAMCO INSTITUTE OF TECHNOLOGY

(Approved by AICTE - New Delhi and Affiliated to Anna University - Chennai)
NAAC Accredited and An ISO 9001:2015 certified institution



RIT/MECH/VAC/2022/02

05.07.2022

To

The Director,

Centre for Academic Courses,

Anna University,

Chennai – 600025.

Respected Sir,

Sub : Requesting approval and permission to offer below proposed Value Added Courses
– V Semester B.E. Mechanical Engineering – Academic Year 2022 - 2023 – reg.
Ref : Affiliated Institutions – Guidelines for conducting Value Added Courses –
Anna University Regulations 2017.

With reference to the above, we have proposed to offer the following value added course to our V semester B.E Mechanical Engineering Students for the academic year 2022 – 2023 (ODD Semester). The syllabus and schedule of the proposed value added course is formulated based on the guidelines AU-R2017.

Sl. No.	Course Code	Course Name	Credits
1	MEVAC1907	Python for Mechanical Engineering Systems	2
2	MEVAC1908	Automation Design of Soft Robots	2

Hereby, we request your kind approval and permission to offer these proposed value added courses to our V semester B.E Mechanical Engineering Students for the academic year 2022 – 2023 (ODD Semester).

Thanking you,

Yours faithfully,


HOD/Mech


Vice-Principal




Principal



RAMCO INSTITUTE OF TECHNOLOGY

(Approved by AICTE - New Delhi and Affiliated to Anna University - Chennai)
NAAC Accredited and An ISO 9001:2015 certified institution



RIT/MECH /VAC/2022/01

02.07.2022

To

The Director,

Centre for Academic Courses,

Anna University,

Chennai - 600025.

Respected Sir,

Sub: Requesting permission to offer below mentioned approved Value Added Course-
V Semester B.E. Mechanical Engineering – Academic Year 2022 - 2023 – reg.
Ref: List of approved Value Added Courses – Letter No: 2525/AU/VA/CAC/2019 on
01.07.2019 and Letter No: 2649/AU/VA/CAC/2021 on 02.09.2021.

With reference to the above, we have proposed to offer the following approved value added courses to our V semester B.E. Mechanical Engineering Students for the academic year 2022 – 2023 (ODD Semester). The syllabus and schedule of the approved value added courses along with approved letter received from Centre for Academic Courses, Anna University, Chennai are enclosed for your kind perusal.

Sl. No.	Course Code	Course Name	Credits
1	MVA006	Green Energy Technologies and Management	2
2	MVA023	IoT and Augmented Reality Applications in Mechanical Engineering	2

Hereby, we request your kind approval and permission to offer these value added courses to our V semester B.E Mechanical Engineering Students for the academic year 2022 – 2023 (ODD Semester).

Thanking you,

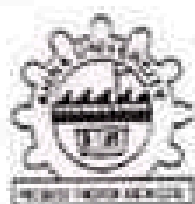
Yours faithfully,


HOD/Mech


Vice Principal




Principal



CENTRE FOR ACADEMIC COURSES
ANNA UNIVERSITY
CHENNAI - 600 025

Off: 22357077 / 73

22357074

Fax / Dir: 22352272



20.07.2022

Dr. S. HOSIMIN THILAGAR
DIRECTOR
Letter No:4190/AU/VA/CAC/2022



To
The Controller of Examinations
Anna University
Chennai - 25.

Sir,

Sub : A.U. - CAC - Non - Autonomous Colleges affiliated to Anna University - Value Added Courses - Reg.

Ref : Letter No.4190/AU/CAC/2022, dated: 13.07.2022.

With reference to the letter cited, the following Value Added Courses offered by Ramco Institute of Technology, Virudhunagar District, Non - Autonomous Colleges affiliated to Anna University are allotted the Course Code as detailed below.

SL. No.	Code Allotted	Title	Credits			
			L	T	P	C
1.	MVA026	Python for Mechanical Engineering Systems	2	0	0	2
2.	MVA027	Automation Design of Soft Robots	2	0	0	2

This is for your kind information and necessary action at your end.

Yours faithfully,

DIRECTOR

Copy to:

1. The Chairperson, Faculty of Mechanical Engineering, A.U., Ch -25.
2. The Principal, Ramco Institute of Technology, North Vengalloor, Ayyanarkovil Road, Rajapalayam - 626 117.
3. The Stock File - CAC.



CENTRE FOR ACADEMIC COURSES

ANNA UNIVERSITY
CHENNAI - 600 025

Off: 22357077 / 73

22357074

Fax / Dir: 22352272



Dr. S. HOSIMIN THILAGAR
DIRECTOR
Letter No.2649/AU/VA/CAC/2021

To
The Controller of Examinations
Anna University
Chennai - 25.

Sir/Madam,

Sub: A.U. - CAC - Affiliated Institutions - Value Added Course - Reg.
Ref: Letter No.RIT/MECH/VAC/2021/04 dated 27.07.2021.

With reference to the letter cited, the following Value Added Course offered by Ramco Institute of Technology, Virudhunagar District, Affiliated Institution is allotted the Course Code as detailed below.

SL No.	Code Allotted	Title	Credits			
			L	T	P	C
1.	MVA023	IoT and Augmented Reality Applications in Mechanical Engineering	1	0	2	2

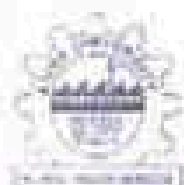
This is for your kind information and necessary action at your end.

Yours faithfully,


DIRECTOR

Copy to:

1. The Chairperson, Faculty of Mechanical Engineering, A.U., Ch -25.
2. The Principal, Ramco Institute of Technology, North Vengannallur, Ayyanarkovil Road, Rajapalayam - 626 117.
3. The Stock File - CAC.



CENTRE FOR ACADEMIC COURSES
ANNA UNIVERSITY
CHENNAI - 600 025

Off: 22357077 / 73
22357074
Fax / Dir: 22352272

Dr. R. RAJU
DIRECTOR

Letter No 2525/AU/VA/CAC/2019

01.07.2019

To
The Controller of Examinations
Anna University
Chennai - 25

Sr.

Sub: A U - CAC - Affiliated Institutions - Value Added Course - Reg.
Ref: Letter No RIT/MECH/VAC/2019/02 dated:20.05.2019

With reference to the letter cited, the following Value Added Course offered by Ramco Institute of Technology, Rajapalayam, Affiliated Institutions is allotted the course code as detailed below

SL.No	Code Allotted	Title
1	MVA005	Smart Materials and Structures
2	MVA006	Green Energy Technologies and Management
3	MVA007	Automation Suite for Smart Systems
4	MVA008	Internet of Things Applications in Mechanical Engineering
5	MVA009	Surface Coating Technology

This is for your kind information and necessary action at your end.

Yours faithfully


DIRECTOR

Cop. to:

1. The Chairperson, Faculty of Mechanical Engineering, A.U. Ch -25
2. The Principal, Ramco Institute of Technology, North Vengalloor, Rajapalayam - 625 117
3. The Stock File - CAC

OBJECTIVES:

- To learn the basics of python programs from an engineering point of view.
- To select an appropriate python library for mechanical relevant problems.
- To know the importance of python in mechanical fields in research and computations.

I PYTHON PROGRAMMING

9

Data types in python, loops – for loop, if loop, while loop, functions in python, Numpy Library – working with arrays/Matrices. Plotting library, Matplotlib: figures and axes, subplots with Matplotlib, Grid specs-Plot Layouts, contour plots, surface plots, polar plots.

II STATIC SYSTEMS

9

Projectile motion, Von Misses failure theory plots, analysis of beams-shear force and bending moment diagrams, kinematic analysis of mechanisms, use of python script in design and simulation software, failure criteria, animating the kinematic mechanism-pendulum animation.

III DYNAMIC SYSTEMS

9

Solving an ordinary differential equation with SciPy ODEINT, analysing the simple spring-mass system with python, single degree of freedom system with SciPy ODEINT, plotting frequency response of the spring-mass system, natural frequency calculation-Eigen value problem, the frequency response of multi-degree freedom system.

IV THERMAL SYSTEMS

9

Plotting and visualising the fluid flow streamlines, analysing the Diesel cycle with python program, one-dimensional heat equation with animated plot, temperature distribution –2D heat transfer problem, flow over cylinder – streamline plot, pressure distribution.

V ROBOTIC SYSTEMS

Interfacing the PIR, Infrared proximity and ultrasonic proximity sensors, - basics of servo motors, basic switching, Projects: Bluetooth-controlled robotic car, autonomous emergency braking system, making a robotic arm, guard robot & Pet feeding robot.

TOTAL: 45 PERIODS**OUTCOMES:**

On successful completion of the course, the student will be able to:

- Use loops, functions, different libraries, and custom functions in python programming.
- Develop programs on static, dynamic, thermal and robotic system concepts.
- Build a project on mechanical engineering domain subject using python.



Assessment Method (Weightage 100 %)			
In Semester	Assessment I	Assessment II	
	MCQ/Short Q & A	MCQ/Simple Experiments	Total
	50%	50%	100%

REFERENCES:

1. Claus Fuhrer, Jan Erik Solem, O. V. (2014). Computing with Python: An Introduction to python for science and engineering. In *Paper Knowledge. Toward a Media History of Documents* (Vol. 7, Issue 2). Pearson Education Limited.
2. Kenan, A. (2020). *Python for Mechanical & Aerospace Engineering* (Vol. 59).
3. Fletcher Heisler, David Amos, Dan Bader, J. J. (2020). Python Basics: A Practical Introduction to Python 3. *Antimicrobial Agents and Chemotherapy*.
<http://www.ncbi.nlm.nih.gov/pubmed/25246403>
4. H, B. (2019). *Python Basics. A self-Teaching Introduction*. Mercury Learning and Information, Dulles, Virginia, Boston, Massachusetts, New Delhi.
5. Ferdinand P.Beer, E.Russel Johnston, Jr., David F. Mazurek, Phillip J.Cornwell, E. R. E. (2007). Vector Mechanics for Engineers: Statics & Dynamics, Ninth Edition. In *Mc Graw Hill Higher Education*. <http://arxiv.org/abs/1011.1669>
6. Joseph, L. (2015). Learning Robotics Using Python. In *Packt Publishing* (Vol. 44, Issue 8).
[https://books.google.com.co/books?hl=es&lr=&id=3ueuQAAQBAJ&oi=fnd&pg=PP1&dq=Lentin,+J.:+Learning+Robotics+Using+Python.+PACKT+\(2015\)&ots=UKvVJB8CmX&sig=XoGsrh-QxQd3bxtLHM2Q91xQK20&redir_esc=y#v=onepage&q&f=false%0Ahttp://arxiv.org/abs/1011.1669%0Ahttp://](https://books.google.com.co/books?hl=es&lr=&id=3ueuQAAQBAJ&oi=fnd&pg=PP1&dq=Lentin,+J.:+Learning+Robotics+Using+Python.+PACKT+(2015)&ots=UKvVJB8CmX&sig=XoGsrh-QxQd3bxtLHM2Q91xQK20&redir_esc=y#v=onepage&q&f=false%0Ahttp://arxiv.org/abs/1011.1669%0Ahttp://)
7. Vaish, D. (2018). Python Robotics Projects. Packt Publishing Ltd.

WEB REFERENCES:

1. <https://www.python.org/>
2. <https://www.coursera.org/specializations/python>
3. <https://www.w3schools.com/python/>
4. <https://www.tutorialspoint.com/python/index.htm>
5. <https://nptel.ac.in/courses/112103109>
6. <https://nptel.ac.in/courses/112103307>
7. <https://nptel.ac.in/courses/112105249>
8. <https://www.aboutmech.com/2021/09/python-for-mechanical-engineers.html>
9. <https://skill-lync.com/mechanical-engineering-courses/python-mechanical-engineers>
10. <https://www.educative.io/courses/python-mech-aero-engineering>


Course Designer


Course Coordinator(s)


HOD/ Mechanical


Vice-Principal


Principal



RAMCO INSTITUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE-ADDED COURSE TIMETABLE

Academic year 2022 - 2023 (Odd Semester)

Title of the Course : MEVAC1907 - PYTHON FOR MECHANICAL ENGINEERING SYSTEMS

Year / Branch / Semester: III / Mechanical / V

Course Handling Faculty

Industrial Experts:

1. Mr.R.Vignesh, Chief Technical Officer, Quantanics Pvt. Ltd. Madurai
2. Mr.G.Guru Prasad , CEO & Founder, Medsby Healthcare and Engineering Solutions, Coimbatore.

Eminent Academicians from other Institutions:

1. Dr.K.Thirumorthy, Assistant Professor (Sr. Grade), Mepco Schlenk Engineering College, Sivakasi
2. Mr.C.Balasubramanian, Assistant Professor (Sr. Grade), Mepco Schlenk Engineering College, Sivakasi

Internal Faculty Members:

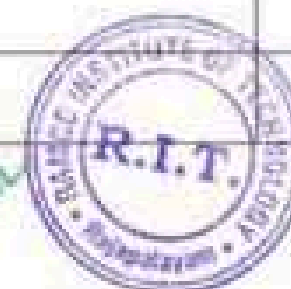
1. Mr.Jerold John Britto, Assistant Professor(Sr. Grade)/ Ramco Institute of Technology
2. Mr.R.Prabhakaran, Assistant Professor/ Ramco Institute of Technology
3. Mr.S.Kathiravan, Assistant Professor/ Ramco Institute of Technology
4. Mr.R.Venkatesh, Assistant Professor/ Ramco Institute of Technology

Evaluation Team Members: 1. Head of the Department

2. One senior faculty member handling the course

3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
1.	06.08.2022 (Saturday)	PYTHON PROGRAMMING - Data types in python, loops – for loop, if loop, while loop, functions in python, Numpy Library – working with arrays/Matrices.	3	-
2.	07.08.2022 (Sunday)	Plotting library, Matplotlib: figures and axes, subplots with Matplotlib, Grid specs-Plot Layouts, contour plots, surface plots, polar plots.	3	-
3.	13.08.2022 (Saturday)	Hands on Training – Python Programming	-	3
4.	14.08.2022 (Sunday)	Hands on Training – Python Programming	-	3



Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
5.	20.08.2022 (Saturday)	STATIC SYSTEMS - Projectile motion, Von Misses failure theory plots, analysis of beams-shear force and bending moment diagrams, kinematic analysis of mechanisms	3	-
6.	21.08.2022 (Sunday)	Hands on Training – Static Systems	-	3
7.	27.08.2022 (Saturday)	Use of python script in design and simulation software, failure criteria, animating the kinematic mechanism-pendulum animation	3	-
8.	28.08.2022 (Sunday)	Hands on Training – Static Systems	-	3
9.	03.09.2022 (Saturday)	DYNAMIC SYSTEMS - Solving an ordinary differential equation with SciPy ODEINT, analysis of the simple spring-mass system with python, single degree of freedom system with SciPy ODEINT	3	-
10.	04.09.2022 (Sunday)	Hands on Training - Dynamic Systems	-	3
Assessment 1	10.09.2022 (Saturday)	The assessment will be conducted as descriptive /MCQ type covers Session 1 to 10		
11.	11.09.2022 (Sunday)	Plotting frequency response of the spring-mass system, natural frequency calculation-Eigen value problem, the frequency response of multi-degree freedom system.	3	-
12.	17.09.2022 (Saturday)	Hands on Training – Dynamic Systems	-	3
13.	18.09.2022 (Sunday)	THERMAL SYSTEMS - Plotting and visualising the fluid flow streamlines, analyzing the Diesel cycle with python program, one-dimensional heat equation with animated plot	3	-
14.	24.09.2022 (Saturday)	Hands on training – Thermal Systems	-	3
15.	25.09.2022 (Sunday)	Temperature distribution –2D heat transfer problem, flow over cylinder – streamline plot, pressure distribution.	3	-
16.	01.10.2022 (Saturday)	Hands on training – Thermal Systems	-	3
17.	15.10.2022 (Saturday)	ROBOTIC SYSTEMS - Interfacing the PIR, Infrared proximity and ultrasonic proximity sensors, - basics of servo motors, basic switching.	-	-

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Sam



Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
18.	16.10.2022 (Sunday)	Hands on training – Robotic Systems	-	3
19.	22.10.2022 (Saturday)	Projects: Bluetooth-controlled robotic car, autonomous emergency braking system, making a robotic arm, guard robot & Pet feeding robot.	3	-
20.	29.10.2022 (Saturday)	Hands on training – Robotic Systems	-	3
Assessment II	30.10.2022 (Sunday)	The assessment will be conducted as practical/descriptive/ MCQ type covers sessions 11 to 20.		
Number of Periods for Theory and Practical Session (Practical sessions will be considered as 50% of the total periods)			30	30
Total number of periods (2 credit courses)			45	

Note: The assessment for the above course will be conducted by considering all the criteria of the rubrics method and will be assessed by the evaluation committee constituted by the Head of the department


Course Designer


Course Coordinator


HoD/Mechanical


Vice-Principal


Principal





RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University
Accredited by NAAC & An ISO 9001: 2015 Certified Institution
NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Mechanical Engineering

Academic Year 2022-2023 (Odd Semester)

Value Added Course Student Name List

MEVAC1907-Python for Mechanical Engineering

<u>Sl.No</u>	<u>Register No.</u>	<u>Name of the Student</u>
1.	953620114002	ALAGARSAMY N
2.	953620114005	BHARATH A C
3.	953620114007	DHAKSHANA MOORTHY S
4.	953620114008	GANESHKUMAR S
5.	953620114027	PRADEESH S
6.	953620114028	PRAKASH RAJ V
7.	953620114032	SENTHURSANJAY R
8.	953620114034	SIVAKUMAR M
9.	953620114305	ARJUN R
10.	953620114306	ARULMARIAAKASH I
11.	953620114308	BALAMUTHUKUMAR
12.	953620114311	DEVAGURU S
13.	953620114312	GANESH R
14.	953620114313	GOPI KRISHNAN M
15.	953620114315	GURU SATHIESH KUMAR S
16.	953620114316	JAYA SURYA P R
17.	953620114318	JOHNSON S
18.	953620114325	KRISHNA V
19.	953620114331	NAVANEETHAKRISHNAN A
20.	953620114339	SRI BALAJI V
21.	953620114343	VENKATESHWARA RAJA N R

R. V. R.
13/07/22

Class Advisor(s)

13/07/22

HOD/Mech

Month: Dec - Jan 23 - Feb 23

CLASS M - Nov - Alp

2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100	
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COURSE MVA026 Python Fe

S. No.	Roll / Reg. No.	Name	Cumulative
1	002	ALAKHAR SAMY A	
2	005	BHARATH A C	
3	007	DHAKSHANA MOORTHY	
4	008	GANESH KUMAR S	
5	027	PRADDEESH S	
6	028	PRADOSH ADITYA V	
7	032	SENTHIL SUNDAR Y R	
8	034	SEVA KUMAR M	
9	305	BROWN R	
10	306	ABUL MAJID ALAKSHI	
11	308	RAJAMUTHUKUMAR	
12	311	DEVA GURU S	
13	312	GANESH R	
14	313	GOPAL KRISHNAN M	
15	315	GURU SATHISH KUMAR	
16	316	JAYA SURYA P J	
17	318	JOHNSON S	
18	325	KRISHNA V	
19	331	NAIDINEETHA KRISHNA	
20	335	SEERAGAJI V	
21	343	VENKATESH KARAN	

No. of observations

Initial of the Faculty Member

Wednesday, February 22, 2012

CLASS Int. English

[illegible]Month:

CLASS

[illegible][illegible]

OBJECTIVES:

- To understand advanced and emerging technologies.
- To know about the key components that makes up an IoT system.
- To gain the knowledge and skills acquired to build a product of IoT system involving prototyping, programming and data analysis.
- To understand where the IoT & Augmented Reality concept fits within the broader mechanical engineering domain and possible future trends.

I IOT TECHNOLOGY**9**

Introduction - scope - Characteristics, Building blocks - overview of IoT components and communication technologies - logical design - controllers - compiler - Functional blocks.

II MODELLING AND SIMULATION OF ROBOTS**9**

Robot configuration - Modular components - Sensors- Actuators - Integration of sensors - Modeling of robot - Programmable peripheral interface- Robot system design and simulation.

III SMART MANUFACTURING**9**

Introduction - Digital Tools, Product Representation - Exchange technologies and standards - Agile Manufacturing Systems -Mass Customization - Automation - Smart Perception - Sensor networks and devices - Smart energy management of manufacturing processes and facilities.

IV APPLICATION OF IOT**9**

Prototype model of PCB-Manufacturing Industry: IoT Car Parking System, build a Robot Car - Mobile Phone using Wifi Direct- Gas pipe leakage detector system-Liquid level monitoring-garbage segregator & bin level indicator-Automatic vehicle accident detection and rescue system-Syringe infusion pump-Smart dustbin.

V AUGMENTED REALITY**9**

Humanistic Intelligence- Mobile and Wearable Computing. Free-roaming AR. Product development: bed level aid for 3D printer, Augmented reality in engineering education, the holographic tool belt, augmented reality light switch for shop floor.

TOTAL: 45 PERIODS**OUTCOMES:**

On successful completion of the course, the student will be able to:

- Use the concepts of Internet of Things for product development.
- Illustrate the hands-on training on modeling and simulation of robot.
- Employ the smart manufacturing system in order to create standards in modern industry
- Demonstrate the IOT based product available in Commercial Manufacturing fields
- Choose proper hardware components in association with Augmented Reality applications

Assessment Method (Weightage 100 %)			
In Semester	Assessment I	Assessment II	
	MCQ/Short Q & A	MCQ/Simple Experiments	Total
	50%	50%	100%

REFERENCES:

1. Waltenegus Dargie, Christian Poellabauer, "Fundamentals of Wireless Sensor Networks: Theory and Practice", Wiley Series on Wireless Communication and Mobile Computing, 2018.
2. Michael Margolis, Arduino Cookbook, Recipes to Begin, Expand, and Enhance Your Projects, second Edition, O'Reilly Media, 2011.
3. Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds), Architecting the Internet of Things, Springer, 2011.
4. Tom Dieck, M. Claudia, Jung, Timothy, Augmented Reality and Virtual Reality, Springer, 2019.
5. Vijay Madisetti, Arshdeep Bahga, "Internet of Things: A Hands-On Approach" ISBN:0996025510, 2014.

WEB REFERENCES:

1. <https://www.udemy.com/course/introduction-to-internet-of-things-iot-using-arduino/>
2. <https://www.edx.org/course/introduction-to-the-internet-of-things-iot?index=product&queryID=b01b6dd1a4e7ef3a89961070d1711c59&position=1>
3. <https://www.udemy.com/course/develop-augmented-reality-book-ar-business-card-with-unity/>


Course Designer & Coordinator


HOD/ Mechanical


Vice-Principal


Principal

RAMCO INSITIUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE ADDED COURSE TIME TABLE

Academic year 2022 - 2023 (Odd Semester)

Title of the Course : MVA023 – IoT and Augmented Reality Applications in Mechanical Engineering

Year / Branch / Semester: III / Mechanical / V

Course Handling Faculty

Industrial Experts:

1. Mr.G.Guru Prasandh , CEO & Founder, Medsby Healthcare and Engineering Solutions, Coimbatore.
2. Mr.S.Mohamed Safiyur Rahman, Software Project Manager, Robert Bosch Engineering and Business Solutions India (RBEI) Pvt. Ltd., Coimbatore.

Eminent Academicians from other Institutions:

1. Dr.K.Thinamoorthy, Assistant Professor (Sr. Grade), Mepco Schlenk Engineering College, Sivakasi
2. Dr.J.Jerald Inico, Assistant Professor, Department of Computer Science, Loyola College, Chennai

Internal Faculty Members:

1. Mr.Jerold John Britto, Assistant Professor(Sr. Grade)/ Ramco Institute of Technology
2. Mr.R.Venkatesh, Assistant Professor/ Ramco Institute of Technology
3. Mr.R.Prabhakaran, Assistant Professor/ Ramco Institute of Technology
4. Mr.M.Ramar, Assistant Professor/ Ramco Institute of Technology.

Evaluation Team Members: 1. Head of the Department

2. One senior faculty member handling the course

3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
1.	06.08.2022 (Saturday)	Introduction – scope - Characteristics - Building blocks	3	-
2.	07.08.2022 (Sunday)	Overview of IoT components and communication technologies - logical design - controllers - compiler - Functional blocks.	3	-
3.	13.08.2022 (Saturday)	Hands on Training - Arduino Uno, Nano, Nodecmu and Ethernet Shield	-	3
4.	14.08.2022 (Sunday)	Hands on Training - Product Development: Smart Lights and Devices.	-	3

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
5.	20.08.2022 (Saturday)	Robot configuration - Modular components – Sensors– Actuators	3	-
6.	21.08.2022 (Sunday)	Hands on Training - Robotics Design using Tinkercad	-	3
7.	27.08.2022 (Saturday)	Integration of sensors – Modeling of robot - Programmable peripheral interface- Robot system design and simulation	3	-
8.	28.08.2022 (Sunday)	Hands on Training – Arduino based Robot Design using Tinkercad	-	3
9.	03.09.2022 (Saturday)	Introduction - Digital Tools, Product Representation - Exchange technologies and standards - Agile Manufacturing Systems	3	-
10.	04.09.2022 (Sunday)	Hands on Training - An online platform to bring Circuit assemblies to life virtually.	-	3
Assessment I	10.09.2022 (Saturday)	The assessment will be conducted as descriptive /MCQ type covers Session 1 to 10		
11.	11.09.2022 (Sunday)	Robotics and Automation (Perception, Manipulation, Mobility, autonomy), Smart Perception – Sensor networks and devices	3	-
12.	17.09.2022 (Saturday)	Hands on Training – Sensors and Motors simulation using Tinker CAD	-	3
13.	18.09.2022 (Sunday)	Mass Customization – Automation - Smart Perception – Sensor networks and devices – Smart energy management of manufacturing processes and facilities	3	-
14.	24.09.2022 (Saturday)	Hands on training – car parking system	-	3
15.	25.09.2022 (Sunday)	liquid level monitoring-garbage segregator & bin level indicator- Automatic vehicle accident detection and rescue system-syringe infusion pump-smart dustbin.	3	-
16.	01.10.2022 (Saturday)	Hands on training – Home Automation and Panic Alarm for the elderly	-	3
17.	15.10.2022 (Saturday)	Humanistic Intelligence- Mobile and Wearable Computing, Free-roaming AR: Wireless, Sensing, and Meta sensing	3	-
18.	16.10.2022 (Sunday)	Hands on training – Usage of augmented / virtual reality tools	-	3
19.	22.10.2022 (Saturday)	Product development: bed level aid for 3D printer, Augmented reality in engineering education, the holographic tool belt, augmented reality light switch for shop floor	3	-

Session	Date/Day	Topics Covered	Theory (periods)	Practical (periods)
20.	29.10.2022 (Saturday)	Hands on training – Virtual AR Tools	-	3
Assessment II	30.10.2022 (Sunday)	The assessment will be conducted as practical / descriptive/ MCQ type covers Session 11 to 20.		
Number of Periods for Theory and Practical Session (Practical session will be consider as 50% of the total periods)			30	30
Total number of periods (2 credit course)			45	

Note: The assessment for the above course will be conducted by considering all the criteria of rubrics method and will be assessed by evaluation committee constituted by the Head of the department


Course Designer & Coordinator


HoD/Mechanical


Vice-Principal


Principal



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Department of Mechanical Engineering

Academic Year 2022-2023 (Odd Semester)

Value Added Course Student Name List

MVA023- IoT and Augmented Reality Applications in Mechanical Engineering

SLNo	Register No.	Name of the Student
1.	953620114006	CHERAN S P
2.	953620114010	GOKUL K
3.	953620114011	GOKUL RAMANAN A
4.	953620114012	GOPINATH D
5.	953620114013	JASWIN SHREE RAAM P J
6.	953620114016	KARAN S
7.	953620114018	KEERTHIVASAN B
8.	953620114019	MANOJ KUMAR A
9.	953620114021	MARI SELVAM C
10.	953620114026	PRADEEP DURAI S
11.	953620114037	VENKADESH R
12.	953620114301	ABDUL AZEEZ S M
13.	953620114302	AHAMED SATHIK USSAIN M
14.	953620114317	JEYA PRAKASH J
15.	953620114319	KALIRAJAKUMAR B
16.	953620114326	KRISHNAKUMAR M S
17.	953620114332	PARTHIBAN A
18.	953620114335	RAM HARI V
19.	953620114336	RAMKUMAR R
20.	953620114340	SUBRAMANIP
21.	953620114342	THIYANESH SAKTHI R

R. V. R.
13/05/22
Class Advisor(s)

[Signature]
13/5/22
HoD/Mech

OBJECTIVES:

At the end of the course, the student is expected to:

- Perform energy audit and identify the opportunities for energy savings
- Understand the various solar technologies and its applications
- Learn other alternate sources of energy and its advancements

UNIT I	ENERGY CONSERVATION AND ENERGY AUDITING	8
Energy Scenario - India and World, Energy Consumption - Environmental Impacts, Energy Conservation, Energy Auditing - Industries and Automobiles, Bureau of Energy Efficiency.		
UNIT II	SOLAR TECHNOLOGIES AND INSTRUMENTATION	14
Solar Energy – Nomenclature, Measuring Instruments, Applications; Ray Tracing; Solar PV Technologies – Efficiency calculation, Waste Management and recycling.		
UNIT III	GREEN BUILDINGS AND RECENT ADVANCEMENTS	8
Green Buildings; Green energy technologies – Other sources of energy, Research focus and recent advancements.		

TOTAL: 30 PERIODS

OUTCOMES:

Upon the completion of this course the students will be able to

1. Conduct energy audit.
2. Demonstrate possible energy saving opportunities in industries and automobiles.
3. Illustrate solar measuring instruments.
4. Compare the performance of solar applications by simulation and experimentation.
5. Describe recent advancements in green energy technologies

Assessment Method (Weightage 100 %)			
In- Semester	Assessment I	Assessment II	Total
	MCQ / Short Q & A	MCQ / Short Q & A	
	50 %	50 %	100 %

REFERENCES:

1. Energy Manager Training Manual (4 Volumes) available at www.energymanagertraining.com, a website administered by Bureau of Energy Efficiency (BEE), a statutory body under Ministry of Power, Government of India, 2004.
2. Sukhatme. S.P., "Solar Energy", Tata McGraw Hill Publishing Company Ltd., New Delhi, 1997.
3. Tiwari. G.N., Solar Energy – "Fundamentals Design, Modelling & Applications", Narosa Publishing House, New Delhi, 2002.


Course Designer & Coordinator


HOD Mechanical


Vice-Principal


Principal

RAMCO INSTITUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE ADDED COURSE TIME TABLE

(Academic year 2022-2023)

Title of the Course : MVA006 - Green Energy Technologies and Management

Year / Branch / Semester : III / Mechanical / V

Course Handling Faculty

Industrial Experts:

1. Mr.R.Nesamani, Joint Director, Petroleum Conservation and Research Association
2. Mr.M.Vivekanandan (Certified Energy Auditor), CEO, TryCAE, Trichy

Eminent Academicians from other institutions:

1. Dr. M.Premalatha, Professor, Department of Energy and Environment, NITT
2. Dr.V.Saravanan, Professor, Department of Electrical and Electronics Engineering, Thiagarajar College of Engineering
3. Mr.R.Vignesh Kumar, Assistant Professor, Department of Mechanical Engineering, National Engineering College, Kovilpatti

Internal Faculty Members :

1. Dr.V.Sivakumar, Associate Professor / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
2. Mr.M.Sivagaminathan alias Balaji, Assistant Professor (Sr.Gr.) / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
3. Mr.M.Ashok Kumar, Assistant Professor (Sr.Gr.) / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam
4. Mr.R.Arun Kumar, Assistant Professor / Mechanical Engineering / Ramco Institute of Technology, Rajapalayam

Evaluation Team Members: 1. Head of the Department
2. One senior faculty member handling the course
3. One senior faculty member nominated by the Head of the Institution

Session	Date/Day	Topics Covered	Theory (Periods)	Practical (Periods)
I	06.08.2022 (Saturday)	Energy Scenario of India and World, Energy Consumption and Environmental Impacts	3	-
II	07.08.2022 (Sunday)	Energy Conservation, Energy Auditing in Industries and Automobiles	3	-
III	13.08.2022 (Saturday)	Bureau of Energy Efficiency, Field visit on Energy Auditing in Electrical and Mechanical Utilities	1	2

Session	Date/Day	Topics Covered	Theory (Periods)	Practical (Periods)
IV	14.08.2022 (Sunday)	Nomenclature of Solar Energy and Measuring Instruments	3	-
V	20.08.2022 (Saturday)	Solar Energy applications and field exposure on solar thermal applications	1	2
VI	21.08.2022 (Sunday)	A Hands on practice – Solar instruments and experimentation on solar thermal applications	-	3
Assessment I	27.08.2022 (Saturday)	Assessment will be conducted either as MCQ / descriptive type		
VII	28.08.2022 (Sunday)	Introduction and A Hands on practice – Ray Tracing Software	1	2
VIII	03.09.2022 (Saturday)	Introduction to Solar PV Technologies and practical exposure to efficiency calculation on different PV panels	2	1
IX	04.09.2022 (Sunday)	Waste Management and recycling of PV Cells	3	-
X	10.09.2022 (Saturday)	Green Buildings, Field visit on Eco-friendly campus at Ramco Institute of Technology	1	2
XI	11.09.2022 (Sunday)	Other green energy technologies	3	-
XII	17.09.2022 (Saturday)	Research focus and recent advancements in Green Energy Technologies	3	-
Assessment II	18.09.2022 (Sunday)	Assessment will be conducted either as MCQ / descriptive type		
Number of Periods for Theory and Practical Session (Practical session will be consider as 50% of the total periods)			24	12
Total number of periods (2 credit course)			30	

Note: The assessment for the above course will be conducted by considering all the criteria of rubrics method and will be assessed by evaluation committee constituted by the Head of the department


Course Designer & Coordinator


HoD/Mechanical


Vice Principal


Principal



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Department of Mechanical Engineering

Academic Year 2022-2023 (Odd Semester)

Value Added Course

MVA006-Green Energy Technology and Management

SLNo	Register No.	Name of the Student
1.	953620114001	ABINESH P
2.	953620114004	BALACHANDRU S
3.	953620114009	GARNEL MARX V
4.	953620114014	JAYAPRAKASH SANKAR M
5.	953620114015	JEEVA M
6.	953620114020	MANOJ KUMAR M
7.	953620114022	MARUTHUPANDI MADHAVAN S
8.	953620114029	PRASANNA GOPI RAM R
9.	953620114031	SAKTHI JEYACHANDRAN M
10.	953620114033	SHASHWATH D
11.	953620114035	SREE VIGNESH N
12.	953620114038	VISHWA M
13.	953620114040	YUVARAJ S K
14.	953620114303	ANBARASU B
15.	953620114307	ASHISH G R
16.	953620114309	BRAHAN MADHAVA SWAMINATH V
17.	953620114321	KAUSHIKRAM R
18.	953620114328	MATHAN J
19.	953620114329	MOHAMED SULFIHER N
20.	953620114334	RAMESH RAJA N
21.	953620114338	SASIKUMAR R

R.vel
12/07/22

Class Advisor(s)

13/5/22
HoD/Mech

OBJECTIVES:

- To learn the basic concepts of virtual robots.
- To know the deep understanding of UiPath tools.
- To design and create robots for business processes independently.
- To develop skills required to attain UiPath RPA Badge.

I INTRODUCTION

09

Automation in Business- Attended and Unattended Robots- Setup with Studio and Assistant- Orchestrator-Building Automation Process- stages of the automation lifecycle- Variables- Arguments- Controlflow in Studio and their types- Applications

II DATA TABLES AND SELECTORS

09

Data Manipulation- Strings, List, and Dictionaries- RegEx Builder- DateTime- DataTables- Workbook activities and Excel App Integration activities- work with Excel files (read data, write data and save files)- Select method- Selectors-Types-UI explorer- Fine Tuning-IMDB Movie Rating-RPA Challenge

III UI AUTOMATION

09

Concepts of UI Automation- modern and the classic design experiences in Studio- Recording UI interactions- Troubleshooting and debugging- Input, output methods and activities-UI synchronization activities-Robot execution logs- Amazon Data Scrapping-Recording Demo

IV PDF AUTOMATION AND DEBUGGING

09

PDF Activities Package-Extracting Data from PDF- Anchor Base- TryCatch, Throw, Rethrow and Retry Scope activities-Global Exception Handler-debug actions, panels and features- Calculate Client Hash

V E-MAIL AUTOMATION AND ORCHESTRATOR

09

Email activities-Retrieve, Filtering and Sending messages- message templates- organizing projects in workflow- Version control system integrations- Orchestrator entities, tenants and folders- Storage buckets, queues, Monitoring and alerts- Capstone Project

TOTAL: 45 PERIODS**OUTCOMES:**

Upon the completion of this course the students will be able to

1. Describe the main stages of the Automation
2. Demonstrate RPA Challenge using selectors
3. Design a UI Automation and verify in data Scrapping applications
4. Extract a single piece of information from a PDF document and explain what all the debug panels indicate.
5. Create and configure unattended robots from Orchestrator.

Assessment Method (Weightage 100%)					
In-semester	Assessment Tool	Capstone Project	Review	MCQ/QUIZ/Viva Voce	Total
	Weightage	70 %	20 %	10 %	100 %


Course Designer & Coordinator


HoD/Mechanical


Vice-principal


Principal



REFERENCES:

1. Adeel Javed, Robotic Process Automation using UiPath StudioX: A Citizen Developer's Guide to Hyperautomation, Apress, 2021
2. Alok Mani Tripathi, Learning Robotic Process Automation, Packt, 2021
3. Nandan Mullakara, Arun Kumar Asokan, Robotic Process Automation Projects: Build real-world RPA solutions, Packt, 2020

WEB REFERENCES:

1. <https://academy.uipath.com/learning-plans>
2. <https://www.coursera.org/specializations/roboticprocessautomation>



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Principal

RAMCO INSITIUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF MECHANICAL ENGINEERING

VALUE ADDED COURSE DETAILS

Title of the Course : AUTOMATION DESIGN OF SOFT ROBOTS

Year / Branch / Semester : III / Mechanical / V

Course Handling Faculty :

Industrial Experts :

1. Mr. Niyaz Ahmed, Technical Program Manager, Uipath Academy.
2. Mr. S. Vishnu Sankar, CEO, Heysoft Sdn Bhd, Malaysia.

Eminent Academicians from other Institutions:

1. Dr. Suneet Gupta, NVIDIA Certified Instructor, Assistant Professor/CSE, Bennett University, Greater Noida
2. Mr. Abhishek Singh, Assistant Professor/ECE, Gyan Ganga Institute of Technology and Sciences, Jabalpur, Madhya Pradesh

Internal Faculty Members :

1. Dr. P. Sureshkumar, Associate Professor/Mechanical, Ramco Institute of Technology, Rajapalayam
2. Mr. S.Valai Ganesh, Assistant Professor/Mechanical, Ramco Institute of Technology, Rajapalayam
3. Dr. J. Jabinth, Assistant Professor/Mechanical, Ramco Institute of Technology, Rajapalayam
4. Mr. T.Selva Sundar, Assistant Professor/Mechanical, Ramco Institute of Technology, Rajapalayam
5. Mr. P. Pavithran, Assistant Professor/Mechanical, Ramco Institute of Technology, Rajapalayam

Evaluation Team Members: 1. Head of the Department

2. Two senior faculty Members Nominated from Head of the Institution

Session	Date/Day	Topics Covered	Theory Sessions	Practical Sessions
I	06.08.2022 (Saturday)	Automation in Business- Attended and Unattended Robots- Setup with Studio and Assistant	3	-
II	07.08.2022 (Sunday)	Orchestrator-Building Automation Process- stages of the automation lifecycle- Variables-Arguments- Controlflow in Studio and their types-	3	-
III	13.08.2022 (Saturday)	Applications- GenericValue, Select, Invoke Workflow file- Hands on with Uipath for Studio	-	3

5/7/22



03/5/22

Session	Date/Day	Topics Covered	Theory Sessions	Practical Sessions
IV	14.08.2022 (Sunday)	Solving Conflicts, Creating and Managing Branches-Hands on using UiPath Studio-II	-	3
V	20.08.2022 (Saturday)	Data Manipulation- Strings, List, and Dictionaries- RegEx Builder- DateTime	3	-
VI	21.08.2022 (Sunday)	DataTablesWorkbook activities and Excel App Integration activities- work with Excel files (read data, write data and save files)	3	-
VII	27.08.2022 (Saturday)	Select method- Selectors-Types-UI explorer- Fine Tuning- Get and sort data, Set data and Highlight WFT type items- Hands on session	-	3
VIII	28.08.2022 (Sunday)	IMDB Movie Rating-RPA Challenge - Hands on session	-	3
IX	03.09.2022 (Saturday)	Concepts of UI Automation- modern and the classic design experiences in Studio	3	-
X	04.09.2022 (Sunday)	Recording UI interactions- Troubleshooting and debugging	3	-
Review I	10.09.2022 (Saturday)	The students are asked to present the selected automation application		
XI	11.09.2022 (Sunday)	Input, output methods and activities-UI synchronization activities-Robot execution logs- Hand on session	-	3
XII	17.09.2022 (Saturday)	Amazon Data Scraping-Recording Demo- Hand on session	-	3
XIII	18.09.2022 (Sunday)	PDF Activities Package-Extracting Data from PDF- Anchor Base	3	-
XIV	24.09.2022 (Saturday)	TryCatch, Throw, Rethrow and Retry Scope activities-Global Exception Handler	3	-
XV	25.09.2022 (Sunday)	Debug actions, panels and features- Hand on session	-	3
XVI	01.10.2022 (Saturday)	Calculate Client Hash- Hand on session	-	3

Signature

Session	Date/Day	Topics Covered	Theory Sessions	Practical Sessions
XVII	15.10.2022 (Saturday)	Email activities-Retrieve, Filtering and Sending messages- message templates	3	-
XVIII	16.10.2022 (Sunday)	Organizing projects in workflow- Version control system integrations	3	-
XIX	22.10.2022 (Saturday)	Orchestrator entities, tenants and folders, Storage buckets, queues, Monitoring and alerts- Hands on session	-	3
XX	29.10.2022 (Saturday)	Sequence, flowchart or state machine, Libraries- Hand on session	-	3
Review II	30.10.2022 (Sunday)	The students will design and demonstrate their Final Capstone project with quiz and all relevant documents		
Total Sessions			30	30
Total Periods			45	

Note : The assessment for the above course will be conducted by considering all the criteria of rubrics method and will be assessed by Head of the department and senior faculty members.


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Department of Mechanical Engineering

Academic Year 2022-2023 (Odd Semester)

Value Added Course

MEVAC1908-Automation Design for Soft Robots

SLNo	Register No.	Name of the Student
1.	953620114003	BALA ABISHEIK J M
2.	953620114017	KARTHIK R
3.	953620114023	MITHRAN R S
4.	953620114024	MOHAN RAJ D
5.	953620114025	MUNEESWARAN V
6.	953620114030	ROHIT R
7.	953620114039	YUVARAJ S
8.	953620114304	ARAVINTHANATHAN M
9.	953620114310	CHANDRA PRAKASH G
10.	953620114314	GOPINATH R
11.	953620114320	KARTHIKEYAN G
12.	953620114322	KEERTHIVASAN M.L
13.	953620114323	KIRUBHA SUBRAMANIAM P
14.	953620114324	KISHORE V
15.	953620114327	LOKESHKUMAR A
16.	953620114330	MUTHUKANI M
17.	953620114333	PRADEEPMOHAN B
18.	953620114337	SANTHANA KRISHNAN J
19.	953620114341	THIRUVANANTHA KRISHNAN S
20.	953620114344	VIGNESH S
21.	953620114345	VIGNESHWARAN A

R. H. S.
12/07/22
Class Advisor(s)

HoD/Mech